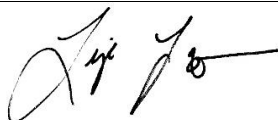


TITLE: Acute Respiratory Distress Syndrome

PROTOCOL: This protocol outlines criteria to diagnose and classify ARDS, provides treatment pathways for Non-Invasive Ventilation and Mechanical Ventilation, and defines criteria for V-V ECMO Consultation.

PURPOSE: To facilitate the early recognition and implementation of appropriate treatments utilizing evidence-based guidelines of ARDS to decrease morbidity and mortality.

APPROVED:

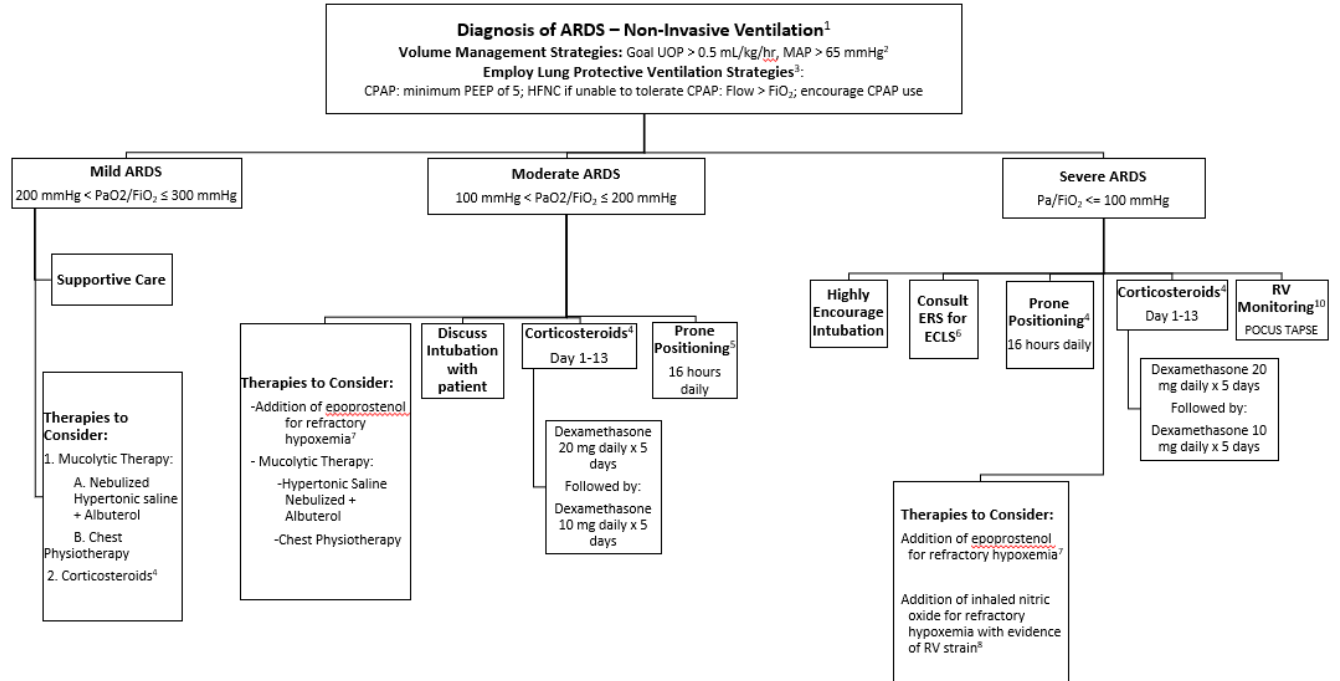


Liza Laquian, MD
Co-Chair ICU Protocol Committee

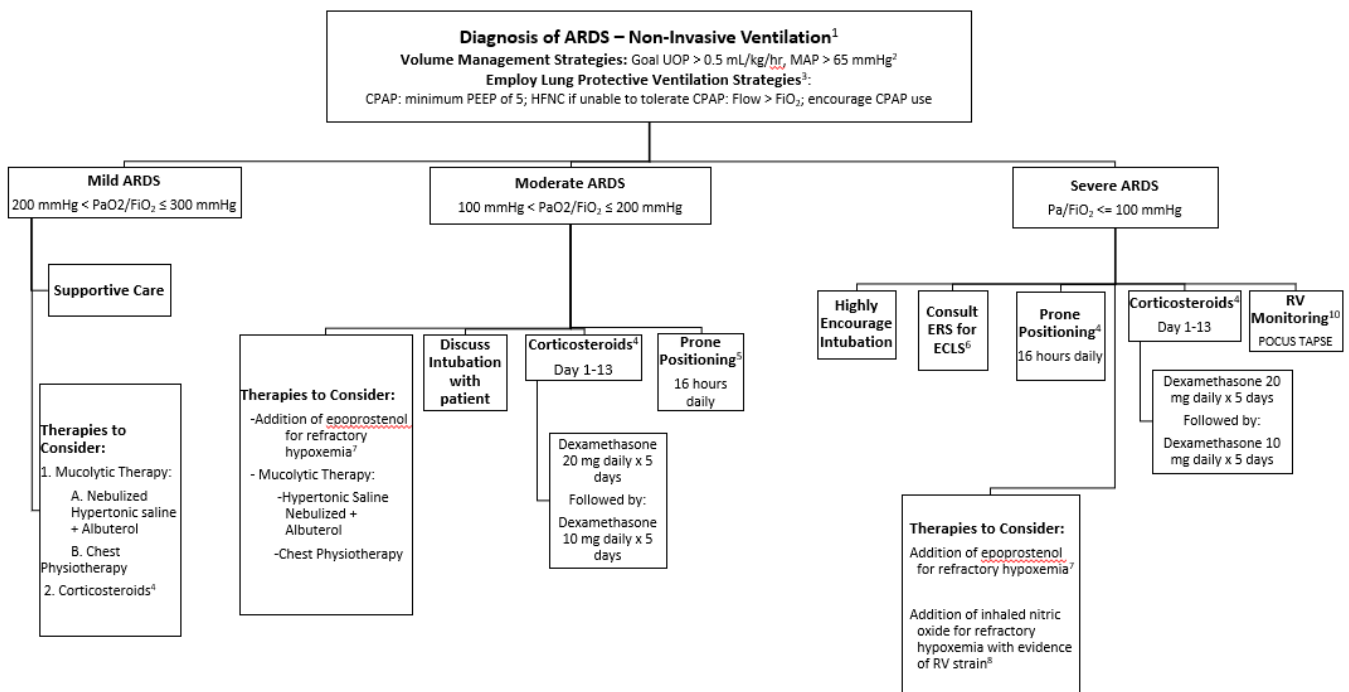


Lindsey Kranson, PA-C
Co-Chair ICU Protocol Committee

NON-INVASIVE VENTILATION ALGORITHM



MECHANICAL VENTILATION ALGORITHM



DEFINITIONS:

- A. ARDS – acute respiratory distress syndrome
- B. POCUS—point of care ultrasound
- C. TAPSE—Tricuspid Annular Plane Excursion
- D. Pplat—plateau pressure
- E. ERS—Emergency Resuscitation Services
- F. THO—Thoracic Surgery
- G. NMB—neuromuscular blockade
- H. NIV—non-invasive ventilation
- I. MV—mechanical ventilation

CORE PROCEDURE:

1. **Diagnostic Criteria for ARDS:**

- a. **Timing:** Within 1 week of known insult
- b. **Radiography:**
 - i. CXR with bilateral infiltrative process
 - ii. CT with 3-4+ affected lobes
 - iii. Lung US – Evidence of bilateral infiltrates (4+ lobes examined)
- c. **Origin of Edema:** Cannot be fully explained by cardiac failure or hypervolemia
- d. **Refractory Hypoxia:**
 - i. Defined:
 - 1. P/F Ratio < 300 or
 - 2. SpO₂ < 90% when FiO₂ > 50% and PEEP ≥ 5 (unless patient on HFNC, Flow > 50)

2. **ARDS Classifications:**

- a. **Mild:** P/F Ratio 201 – 300
- b. **Moderate:** P/F Ratio 101 – 200
- c. **Severe:** P/F Ratio ≤ 100

3. **Volume Management Strategy:**

- a. Individualized
- b. EVEN to NET Negative Balance as hemodynamics allow
- c. Monitor Renal Function and Ensure Adequate End-Organ Perfusion
- d. Utilize daily POCUS to guide fluid management with careful evaluation of RV function (e.g., TAPSE)

4. **Lung Protective Ventilation – Mechanical Ventilation**

a. ARDSNET – Lung Protective Ventilation

- i. Utilize Ideal Body Weight (IBW)
 1. Males = $50 + 2.3 [\text{height (inches)} - 60]$
 2. Females = $45.5 + 2.3 [\text{height (inches)} - 60]$
- ii. Select any Ventilator Mode – for most patients, a pressure-controlled mode with 100% supported breaths will be most appropriate
- iii. Set Ventilator settings to achieve initial $V_t = 6 \text{ mL/kg PBW}$
- iv. If there is respiratory acidosis, start with initial $V_t = 8 \text{ mL/kg PBW}$ and reduce V_t by 1 mL/kg at intervals ≤ 2 hours until $V_t = 6 \text{ mL/kg PBW}$
- v. Set initial rate to approximate baseline minute ventilation (not $> 35 \text{ bpm}$)
- vi. Adjust V_t and RR to achieve pH and plateau pressure goals below

b. Oxygenation Goals

- i. PaO_2 55-80 mmHg or SpO_2 88 – 95%
- ii. Use a minimum PEEP of $5 \text{ cm H}_2\text{O}$
- iii. Consider use of incremental FiO_2 /PEEP combinations as shown below to achieve goal

c. Plateau Pressure Goal: $\leq 30 \text{ cm H}_2\text{O}$

- i. Check Pplat (0.5 sec inspiratory pause), at least q4h and after each change in PEEP or V_t
- ii. If $\text{Pplat} > 30 \text{ cm H}_2\text{O}$: decrease $V_t < 6 \text{ mL/kg}$, increase V_t by 1 mL/kg until $\text{Pplat} > 25 \text{ cm H}_2\text{O}$ or $V_t < 6 \text{ mL/kg}$
- iii. If $\text{Pplat} < 30$ and breath stacking or dyssynchrony occurs: may increase V_t in 1 mL/kg increments to 7 or 8 mL/kg if Pplat remains $\geq 30 \text{ cm H}_2\text{O}$
- iv. pH Goal: 7.2 – 7.3

d. Acidosis Management: (pH < 7.20)

- i. If pH 7.10 – 7.20: Increase RR until pH > 7.30 or $\text{PaCO}_2 < 25$ (Maximum set RR = 35)
- ii. If pH < 7.10 : Increase RR to 35
- iii. Pplat target of 20 may be exceeded
- iv. May give NaHCO_3
- v. Consult to ERS/THO teams

e. Alkalosis Management (pH > 7.45)

- i. Decrease Ventilator Rate if possible
- ii. Consider increasing sedation or use of Neuromuscular Blockade
- iii. I:E Ratio Goal: Recommend that duration of inspiration be $\leq$ duration of expiration

- f. Use of APRV is at Critical Care Attending discretion

5. Corticosteroids

- a. Avoid in Influenza

- b. **Dexamethasone for ARDS (tapering doses)**
 - i. dexamethasone (Decadron) tablet
 - a. 20 mg, oral, every 24 hours, for 5 days
 - i. *followed by*
 - 6. g, oral, every 24 hours, for 5 days
 - 1. **OR**
 - ii. dexamethasone (Decadron) liquid
 - a. 20 mg, oral, every 24 hours, for 5 days
 - i. *followed by*
 - 7. g, oral, every 24 hours, for 5 days
 - 1. **OR**
 - ii. dexamethasone (Decadron) IV
 - a. 20 mg, oral, every 24 hours, for 5 days
 - 2. *followed by*
 - a. 10 mg, oral, every 24 hours, for 5 days
 - b. Notify Surgical team or Burn team (as applicable) of initiation of steroids
 - c. Ensure PPI daily is ordered throughout duration of therapy
8. **Prone Positioning**
- a. Early initiation encouraged (≤ 48 hours of ARDS recognition)
 - b. May be utilized for NIV or MV patients
 - c. Duration ≥ 16 hrs/day
 - d. Refer to “Shands Pronation and Positioning Guidelines for ARDS Patients” on the Bridge
 - e. Prone positioning should not be utilized for ECMO patients
 - f. Absolute Contraindications:
 - i. Sustained ICP > 30 mmHg or CPP < 60 mmHg
 - ii. Unstable Spine or Pelvic Fracture
 - iii. Cervical Fracture
 - iv. Critical Airway Precautions in place
 - v. Transsphenoidal Resection of Pituitary Tumor (TSRPT)
 - vi. Sinus Precautions
 - vii. Orbital Fractures
 - viii. Hemisectomy or Suboccipital Craniectomy
 - g. Relative Contraindications:
 - i. Hemodynamic Instability
 - ii. Known critical/difficult airway
 - iii. Open Abdomen / Chest or recent Sternotomy/Thoracotomy
 - iv. Pregnancy
 - v. Criteria for Immediate Discontinuation of Prone Positioning

- vi. Hypotension refractory to medical therapy (MAP < 60 mmHg, sustained SBP less than 60)
- vii. Inadequate Respiratory Mechanics, Obstruction of ETT, Loss of Airway

9. ECMO Consult

- a. Consult ERS / THO if patient meets one of the following criteria:
 - i. P/F Ratio <80 with FiO₂ >80% for >6 hours despite use of optimized ventilator strategy and PEEP >10
 - ii. Murray Score of ≥ 3
 - iii. pH <7.25 due to hypercarbia despite ventilator optimization
 - iv. ECMO consultation per team discretion
- b. Obtain the following when consulting ERS / ECMO
 - i. CXR
 - ii. ABG
 - iii. CBC
 - iv. TEG
 - v. Fibrinogen
 - vi. PT/INR
 - vii. aPTT
 - viii. Bilateral UE and LE DVT Scans
 - ix. STAT Limited TTE for LV/RV Evaluation
- c. The decision to cannulate will trigger ECMO algorithm

10. Initiation of Inhaled Epoprostenol

- a. Criteria for Use:
 - i. PaO₂ / FiO₂ ratio less than 200 despite FiO₂ $\geq 60\%$ and PEEP ≥ 12 cm H₂O
 - ii. Persistent Critical Hypoxia with maximal ventilator management
- b. Absolute Contraindications:
 - i. Pregnancy
 - ii. Relative Contraindications:
 - iii. Hypersensitivity to drug or dilutants
 - iv. Severe LV Dysfunction
 - v. Thrombocytopenia (platelet count <50,000)
 - vi. Active Significant Bleeding
- c. Discontinue this product if the following occurs:
 - i. Worsening thrombocytopenia with coagulopathic bleeding
 - ii. New LV Dysfunction and worsening pulmonary edema
 - iii. Patient placed on V-V ECMO support
- d. Providers currently pregnant should avoid entering patient room while this product is being administered.

11. Initiation of Inhaled Nitric Oxide

- a. Criteria for Use:
 - i. $\text{PaO}_2 / \text{FiO}_2$ ratio less than 200 despite $\text{FiO}_2 \geq 60\%$ and $\text{PEEP} \geq 12 \text{ cm H}_2\text{O}$
 - ii. Persistent Critical Hypoxia with maximal ventilator management
 - iii. Patient with:
 - a. Refractory hypoxemia despite epoprostenol initiation
 - 2. or
 - a. Refractory hypoxemia with new RV Dysfunction
 - 3. or
 - a. Refractory hypoxemia and is currently pregnant
 - 4. or
 - a. Pre-existing PHTN
- b. Absolute Contraindications:
 - i. Severe LV Dysfunction
 - ii. Congenital Heart Disease involving a right to left shunt
- c. Relative Contraindications:
 - i. Hypersensitivity to drug or diluents
 - ii. CHF secondary to Severe LV Dysfunction
 - iii. Thrombocytopenia (platelet count $< 50,000$)
 - iv. Active Significant Bleeding
 - v. Discontinue this product if the following occurs:
 - vi. Worsening thrombocytopenia with coagulopathic bleeding
 - vii. New LV Dysfunction and worsening pulmonary edema
 - viii. Patient placed on V-V ECMO support and does not have RV Dysfunction

12. Neuromuscular Blockade

- a. Prior to utilization of NMB, utilize different modes of ventilation to optimize compliance
- b. Prior to Initiation of NMB, ensure sedation and analgesia infusion rates dosed to consistent level of:
 - i. RASS goal: -4 to -5
 - ii. ANVPS goal: 0
- c. First Line Agent: Cisatracurium
- d. Neuromuscular Blocking “holidays” shall be performed once daily to ensure sedation and analgesia goals are met
 - i. 60-minute holidays with 4/4 twitches achieved
 - ii. Vecuronium bolus should be immediately available during holiday in case of ventilator dyssynchrony and refractory hypoxia
 - iii. During NMB holiday, if patient does not experience worsening hypoxia to less than 88%, NMB should be discontinued
- e. During NMB, measures should be taken to avoid adverse effects:

- i. Corneal Ulceration
 - 1. Artificial Tear Ointment q4h scheduled with PRN drops
 - 2. Eyelids should be taped
 - 3. Daily examination for infection, injection on conjunctiva
 - 4. Evaluation of Fungal Ophthalmitis if Fungemia occurs
- ii. Critical Illness Myopathy / Weakness
 - 1. Daily assessments to discontinue therapy
 - 2. Passive range of motion exercises
- iii. ICU Delirium
 - 1. Non-Pharmacologic Measure protocol should be in place
 - 2. Sleep/Wake cycles should be highly emphasized. Consider use of Melatonin nightly.
- f. Team should evaluate daily for discontinuation
- g. NMB should be discontinued with ECMO Cannulation

13. Daily POCUS Evaluation

- a. POCUS should include
 - i. Lung Evaluation for pleural effusions and pneumothoraxes
 - ii. Parasternal Long Axis view
 - iii. Parasternal Short Axis view
 - iv. 4-Chamber view with TAPSE calculation
 - v. IVC view
- b. Consider ordering STAT Limited TTE for RV evaluation if new TAPSE <1.7 cm or worsening TAPSE

ASSOCIATED POLICIES:

- 1. Trauma ECMO Protocol
- 2. Neuromuscular Blockade Protocol
- 3. Shands Positioning and Pronation Guidelines for ARDS Patients

KEYWORDS:

ARDS, Acute Respiratory Distress Syndrome, Flolan, inhaled Nitric Oxide, Prone Positioning, ECMO, Neuromuscular Blockade, Steroids, Lung Protective Ventilation

REFERENCES:

- Grasselli G, Calfee CS , et al. European Society of Intensive Care Medicine Taskforce on ARDS. ESICM guidelines on acute respiratory distress syndrome: definition, phenotyping and respiratory support strategies. *Intensive Care Med.* 2023 Jul;49(7):727-759. doi: 10.1007/s00134-023-07050-7. Epub 2023 Jun 16. PMID: 37326646; PMCID: PMC10354163.
- The ARDS Definition Task Force*. Acute Respiratory Distress Syndrome: The Berlin Definition. *JAMA.* 2012;307(23):2526–2533. doi:10.1001/jama.2012.5669
- Lazzeri C, Peris A. The Kigali modification of the berlin definition: a new epidemiological tool for ARDS? *J Thorac Dis.* 2016 Jun;8(6):E443-5. doi: 10.21037/jtd.2016.03.84. PMID: 27294247; PMCID: PMC4886026.
- Acute Respiratory Distress Syndrome Network; Brower RG, Matthay MA, Morris A, Schoenfeld D, Thompson BT, Wheeler A. Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. *N Engl J Med.* 2000 May 4;342(18):1301-8. doi: 10.1056/NEJM200005043421801. PMID: 10793162.
- Villar J, Ferrando C , et al. Dexamethasone in ARDS network. Dexamethasone treatment for the acute respiratory distress syndrome: a multicentre, randomised controlled trial. *Lancet Respir Med.* 2020 Mar;8(3):267-276. doi: 10.1016/S2213-2600(19)30417-5. Epub 2020 Feb 7. PMID: 32043986.
- Chaudhuri D, Nei AM, Rochweg B, et al. 2024 focused update: guidelines on use of corticosteroids in sepsis, acute respiratory distress syndrome, and community-acquired pneumonia. *Crit Care Med.* Online special article. January 19, 2024. doi: 10.1097/CCM.0000000000006172.
- Guérin C, Reignier J , et al. PROSEVA Study Group. Prone positioning in severe acute respiratory distress syndrome. *N Engl J Med.* 2013 Jun 6;368(23):2159-68. doi: 10.1056/NEJMoa1214103. Epub 2013 May 20. PMID: 23688302.
- Schmidt M, Hajage D , et al. PRONECMO Investigators, the REVA Network, and the International ECMO Network (ECMONet). Prone Positioning During Extracorporeal Membrane Oxygenation in Patients With Severe ARDS: The PRONECMO Randomized Clinical Trial. *JAMA.* 2023 Dec 26;330(24):2343-2353. doi: 10.1001/jama.2023.24491. Erratum in: *JAMA.* 2024 Feb 6;331(5):446. doi: 10.1001/jama.2024.0231. PMID: 38038395; PMCID: PMC10692949.